Comparing Current Housing Prices to Expected Housing Prices in Minneapolis/St.Paul

Ben Peters
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Agenda

- Goal/Reason for Modeling
- Available Data
- Plan for Data Workup and Modeling
- Data Cleanup Strategy
- Exploratory Data Analysis
- Summary of Modeling Outcomes
- Final Conclusions
- Next Steps

Predicting Minneapolis/St.Paul Housing Data: Why Do it?

- For the general home buyer, why a price is set the way it is may not make sense
  - Why is one home much more expensive than the other?

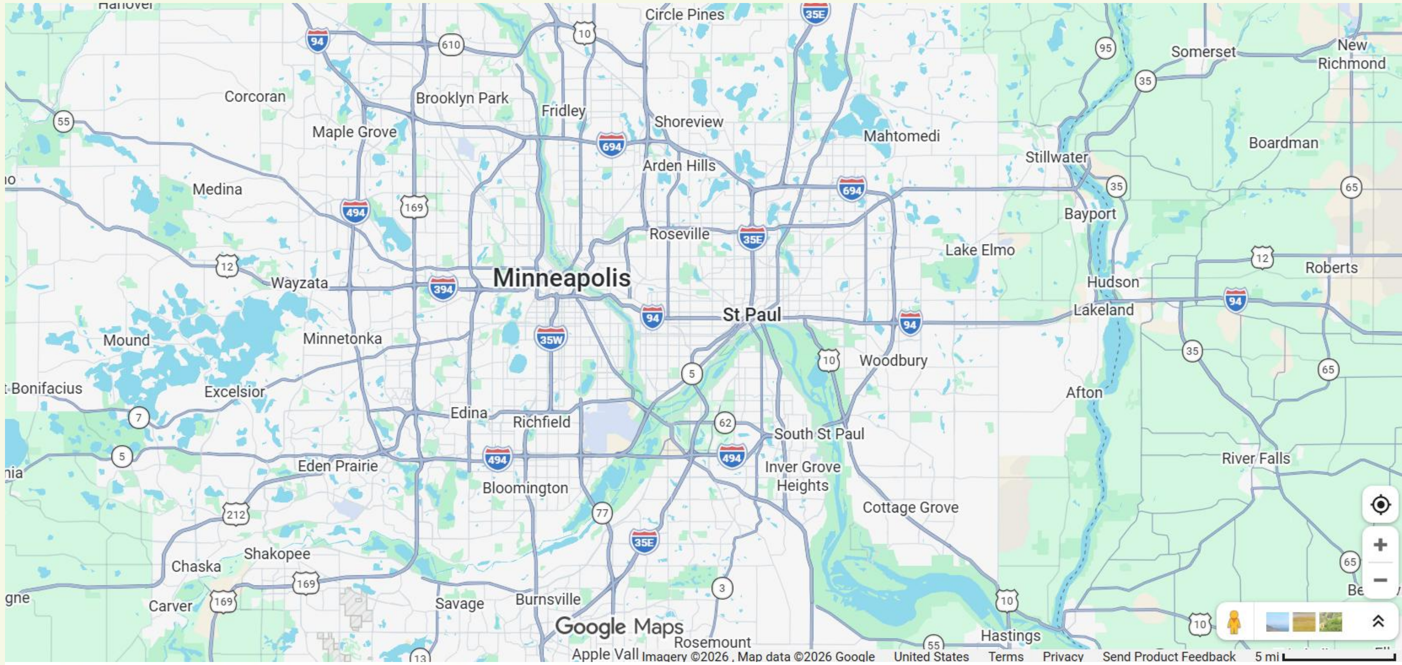
- This may lead the home buyer to pay way too much for a house, or miss an incredible deal

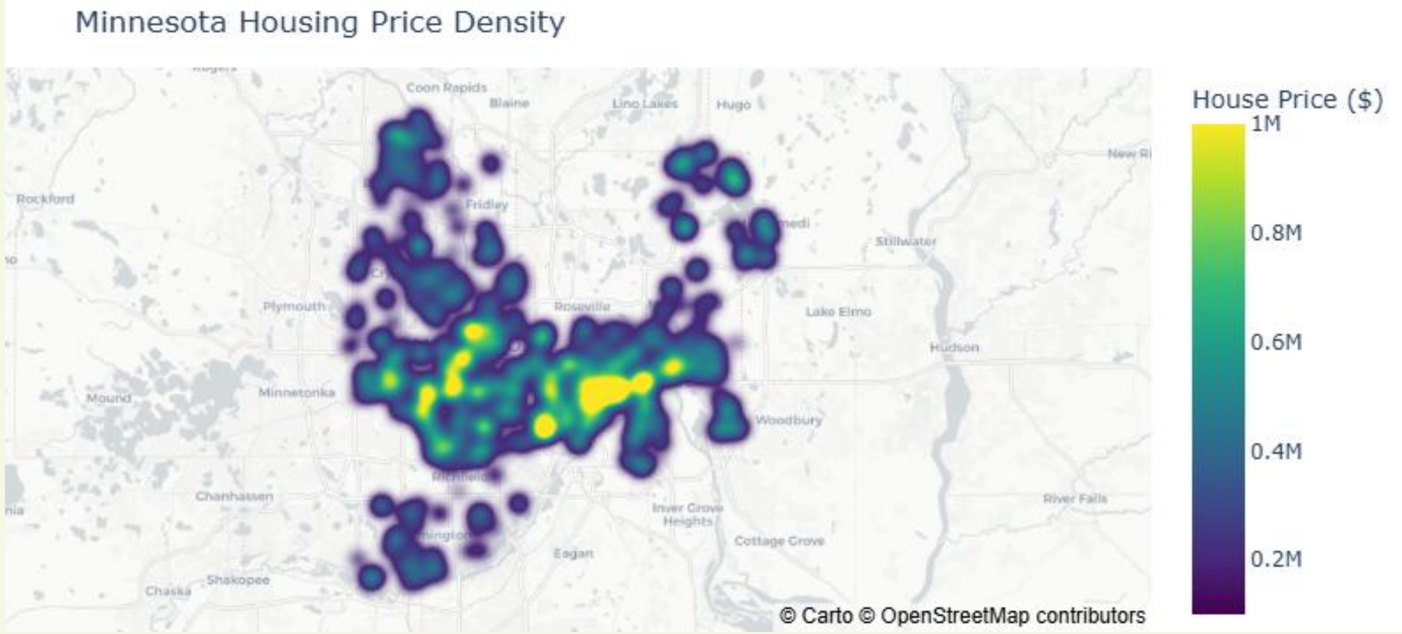
- In order for the homebuyer to have a better idea of housing costs, a model could be used to predict the cost, and then the home buyer could compare to the expected price given by the model

Image from: [Minneapolis Housing Market: Trends and Forecast 2025-2026]

Data Overview:

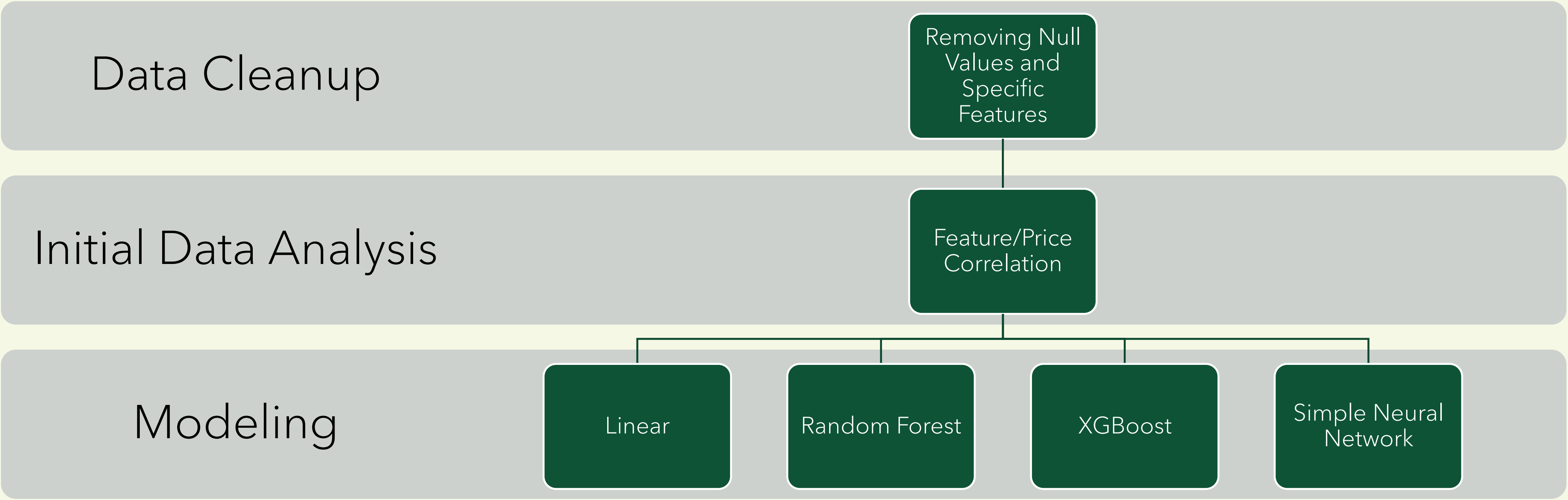
- All houses were on the market at the time the data was acquired
- Initial dataset had 702 properties sourced from Kaggle/Redfin
- All properties were from the Minneapolis/ St. Paul area
    - Limiting the scope allows for less effects from differing laws, taxes, and other data that was not readily available





Data Sourced From: 🏠 United States - Redfin Housing Market (CSV)

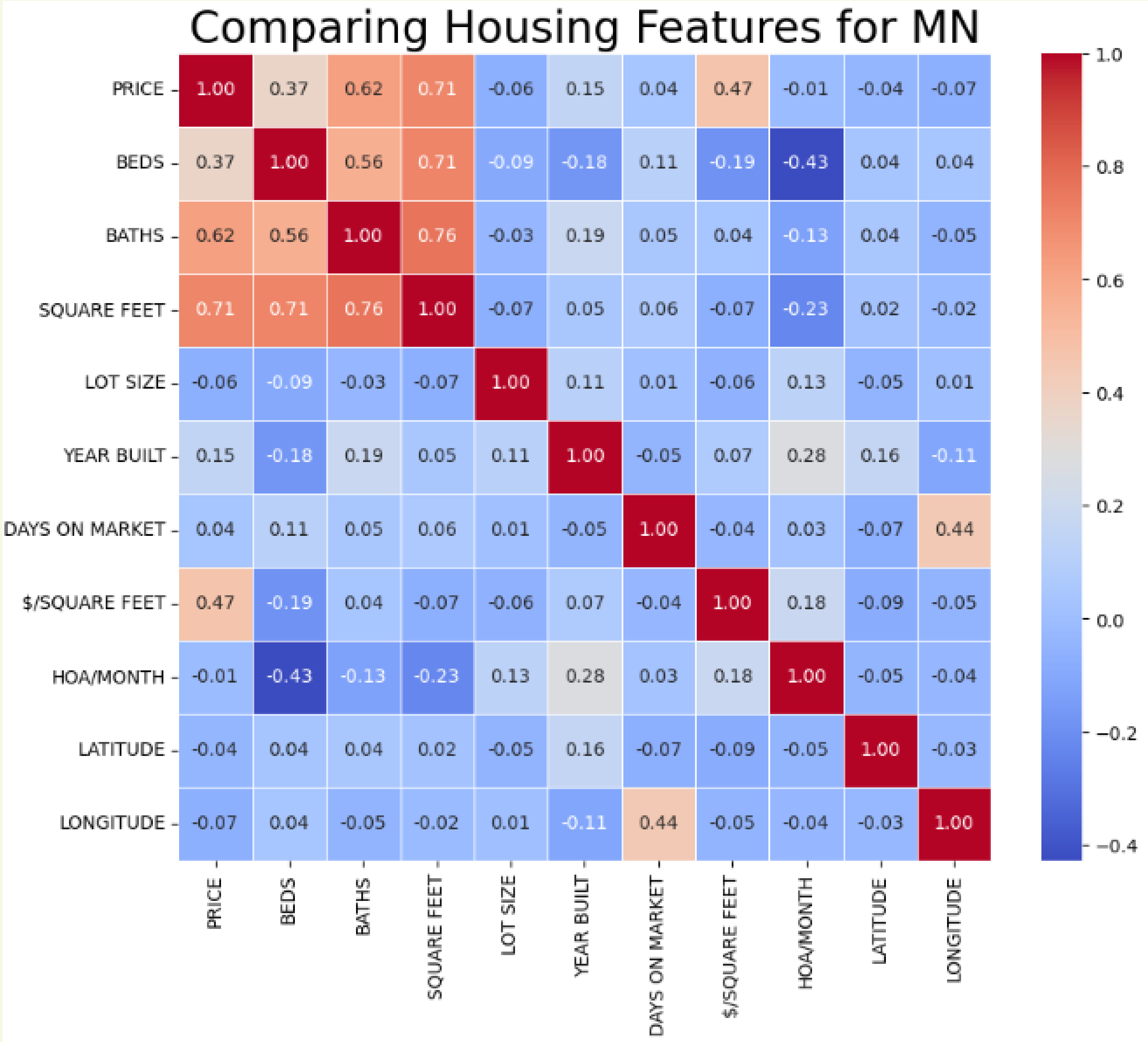
Plan for Data Cleanup and Modeling



- Notes on Data Cleanup for Data Points:
  - 31 total data points were removed because they were vacant land
    - Remaining 671 samples were used for modeling/testing
- Notes on Data Cleanup for Features:
  - 11 features were removed
    - Features included information that was not useful for modeling such as state (all samples are in MN), URL, Sold Date (all houses were on market)

Feature Correlation with Housing Prices

- All numerical features were compared to Price to determine if there are any correlations
- Beds, Baths, Square feet and Price/Square foot all exhibited a positive correlation
- Likely multicollinearity
- Next step: Explore the relationships of these features with Price
- Interestingly, the # Beds had a loose negative correlation with HOA/Month
    - Likely smaller residences such as condos



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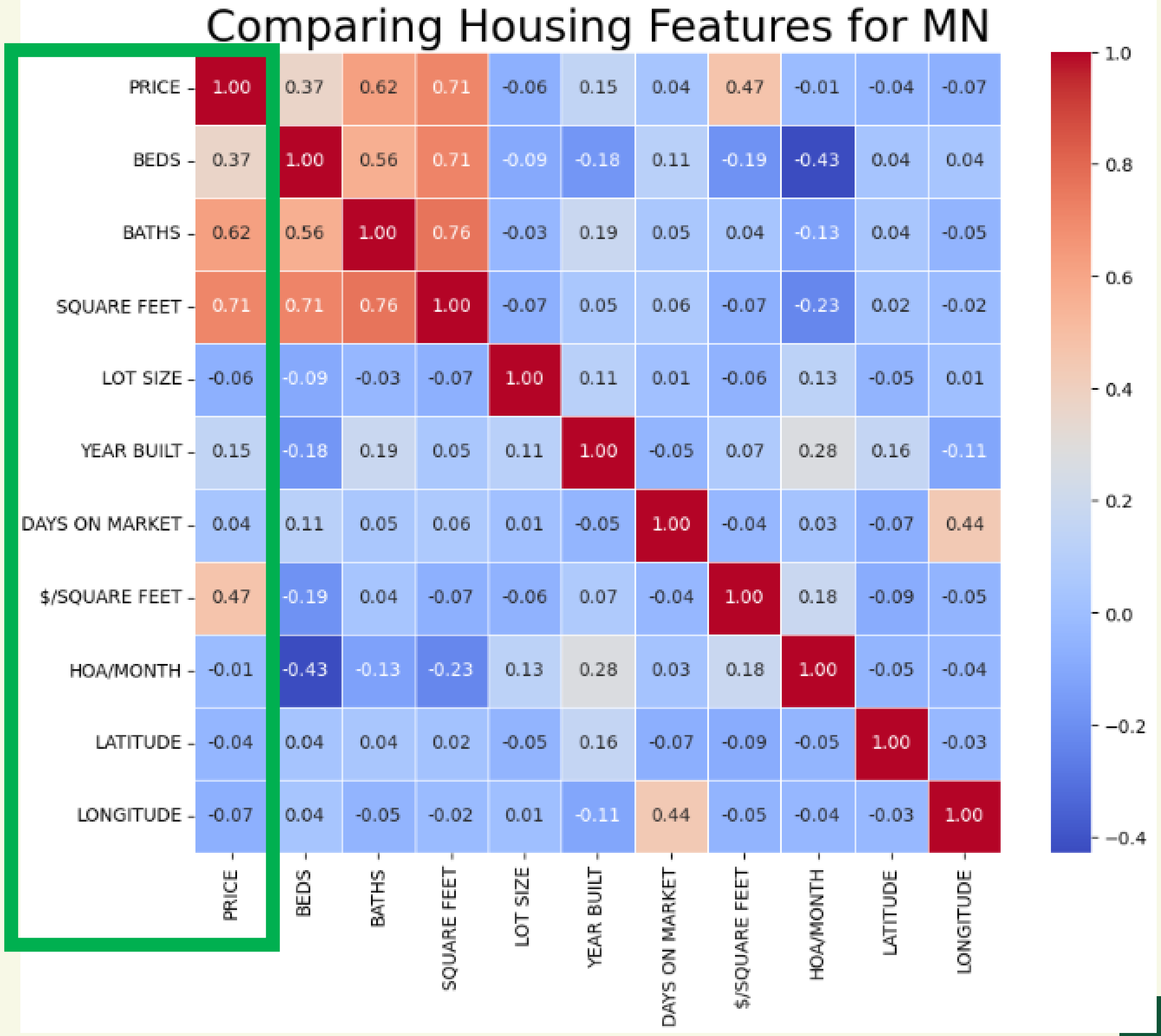
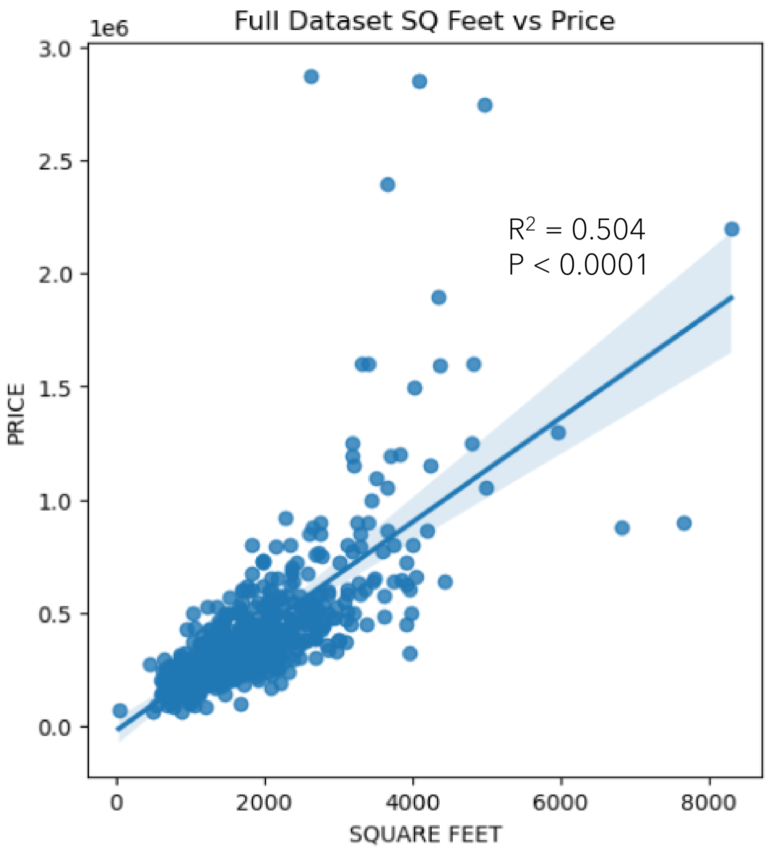
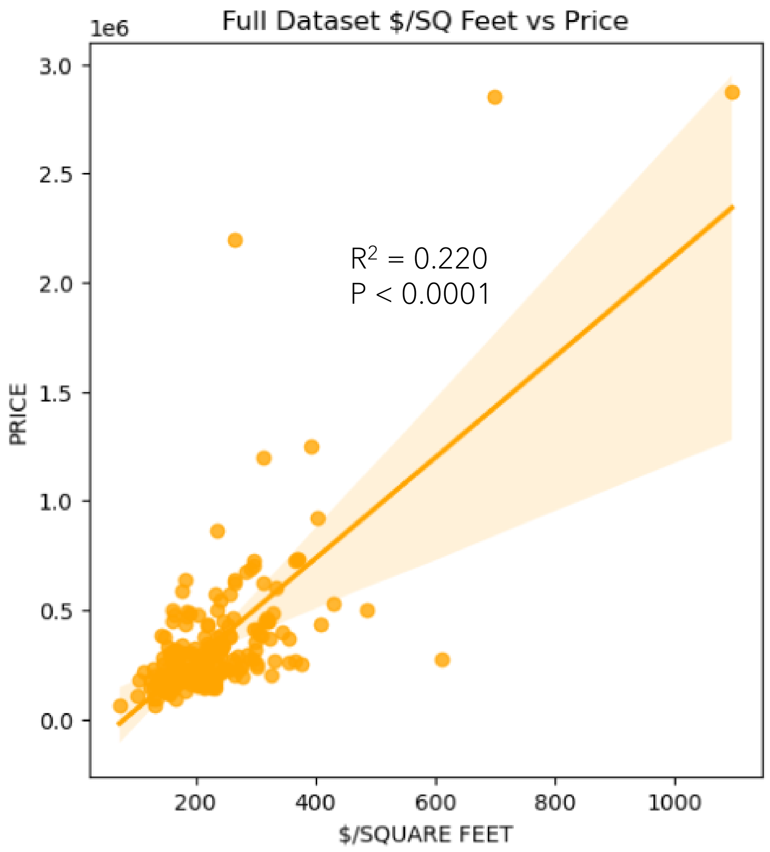


Feature Correlation with Housing Prices: Square Feet

- Square feet has a weak correlation with price
  - This is expected as intuitively larger housing units cost more to build and are in higher demand

- A similar conclusion can be made for Price per square foot
  - Feature removed for modeling purposes





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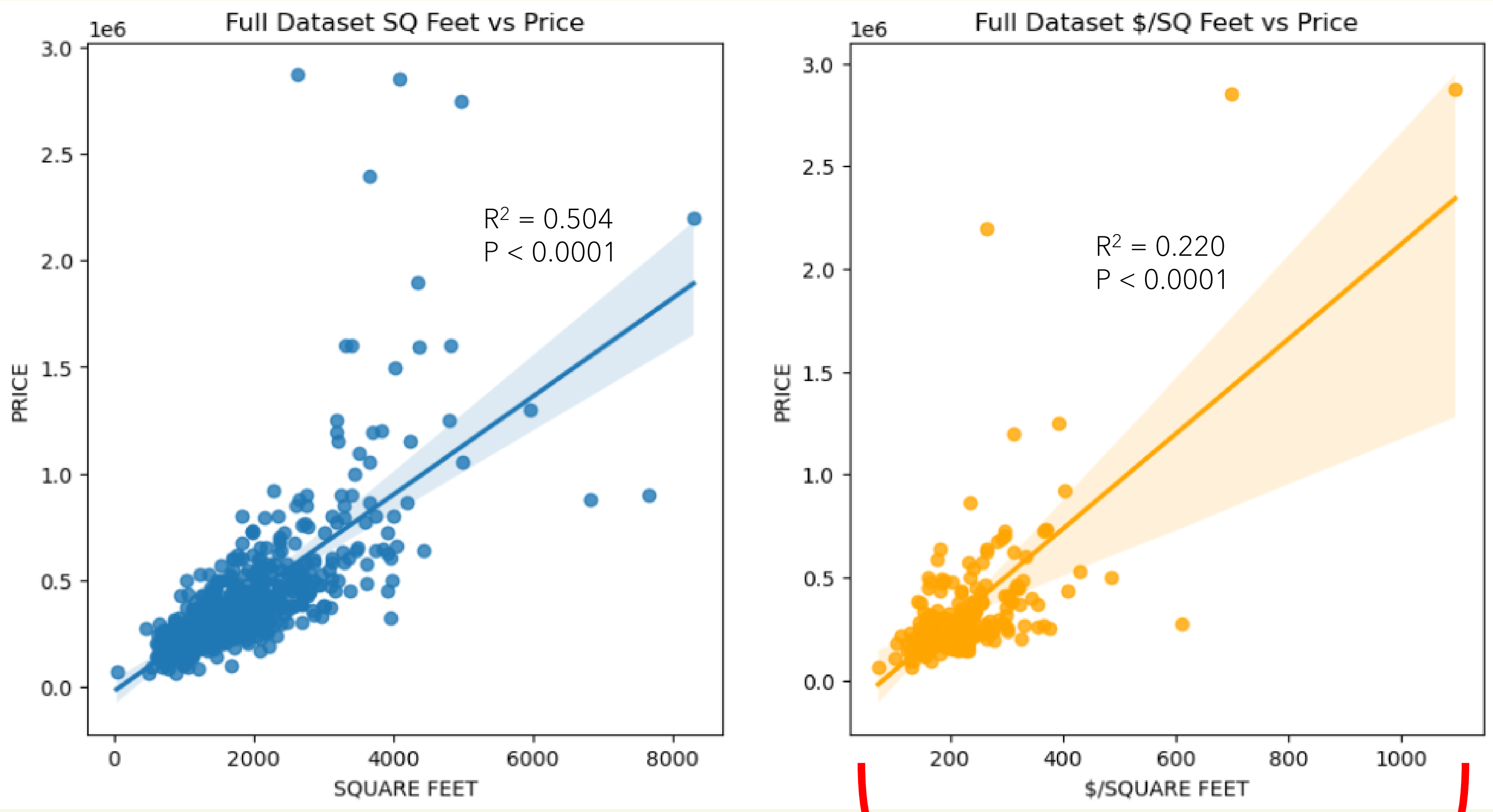


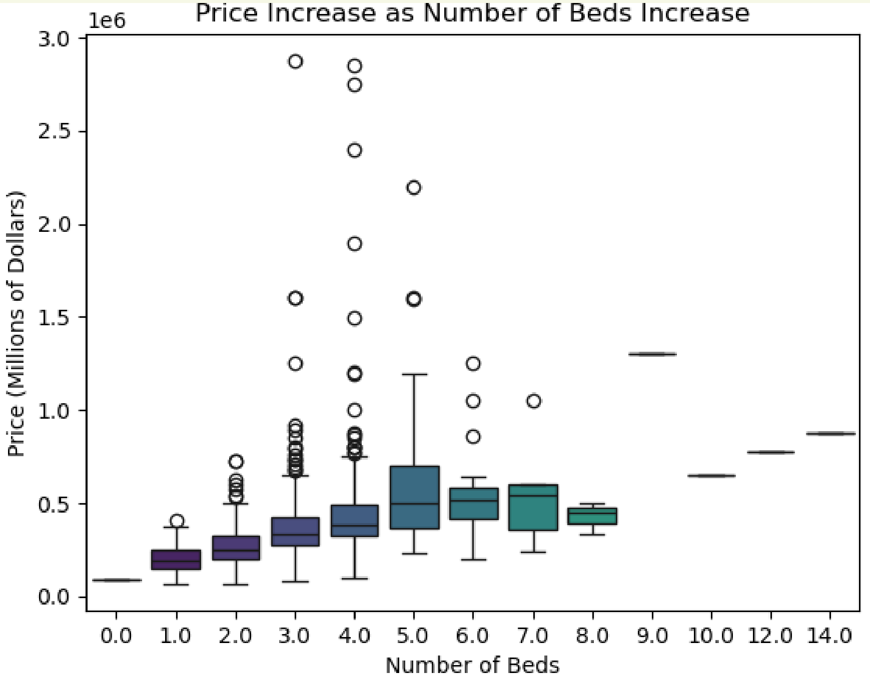
Feature Removed to Prevent Data Leakage

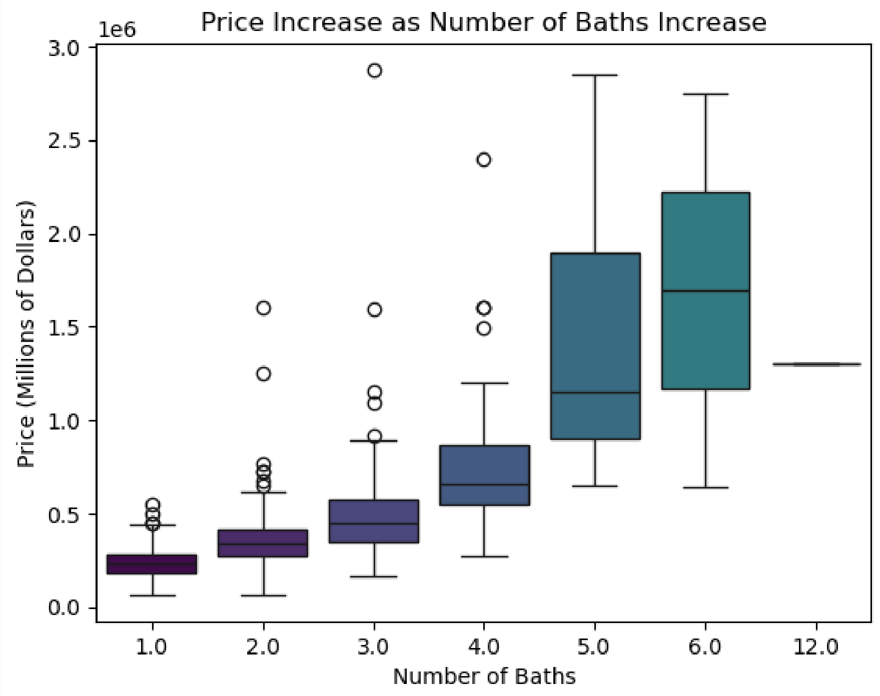
Feature Correlation with Housing Prices: Beds/Baths

- Generally, As the number of beds and baths increase, the price increases as well
  - Price does not seem to increase after 5 beds (very large housing unit

- Through a Pairwise Tukey significance test the price was found to be significantly different for groups of <3 beds, 3-5 beds, and >5 beds ($p < 0.05$)

- Also using the pairwise Tukey significance test, groups of <3 baths, 3-4 baths, and >4 baths were all found to be significant ($p < 0.05$)

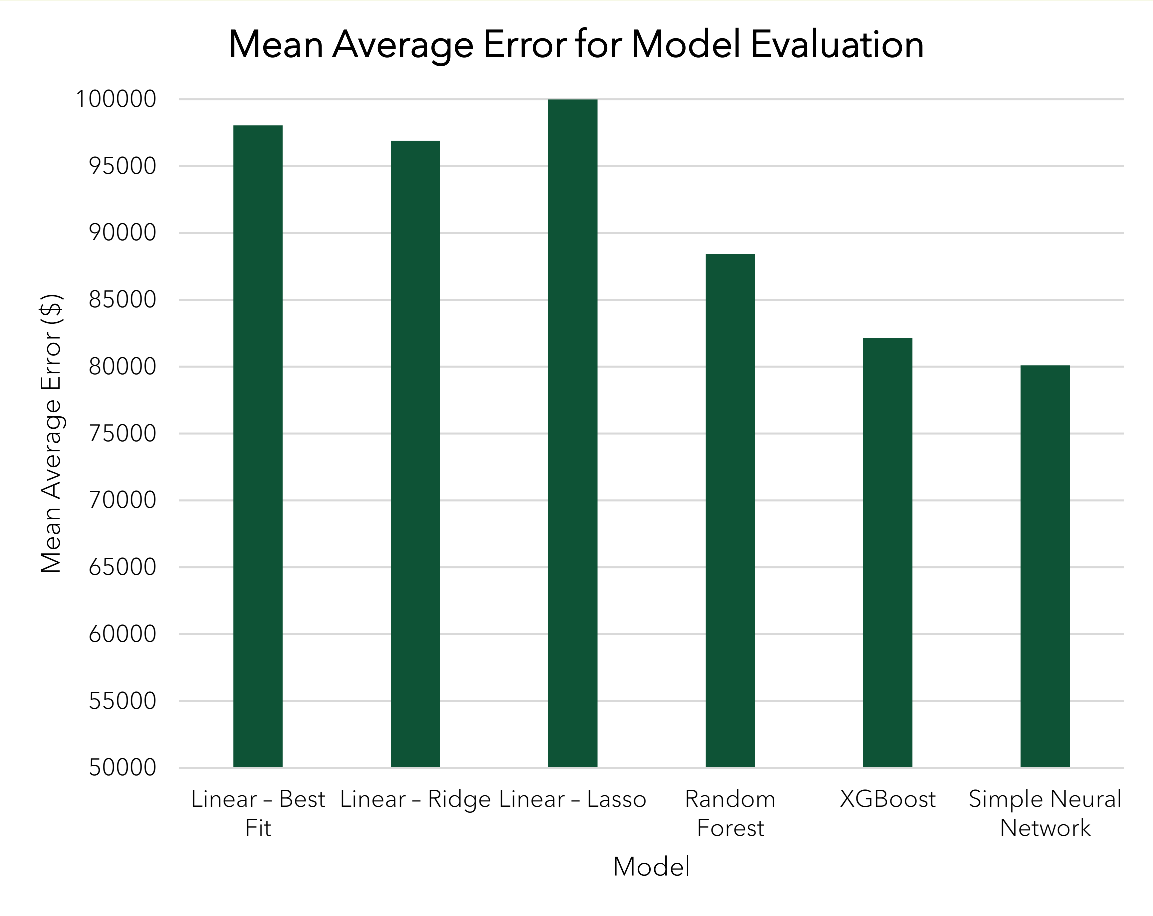




Modeling Summary: Model Optimization

- All models used the same data split (80% training, 20% testing)
- Random Forest and XGBoost models were optimized using a randomize cross validation parameter search to optimize model parameters
    - The models were then further optimized using gridsearch as long as the test MAE did not increase (preventing overfitting)

- Neural Network parameters such as learning rate and # of epochs were modified and optimized to obtain the best fit possible



Modeling Summary: Choosing the Best Model

- To understand the model limitations further, the Mean Absolute Percentage Error was determined for different pricing bins
- XGBoost and Neural Network exhibited the lowest error on the test dataset

Model	R^2	MAE	Low Price (<$300K)	Med Price ($300K-$500K)	Med-High Price ($500K-$1M)	High Price (>$1M)
# of samples	135	135	61	49	20	5
Linear – Best Fit	0.66	$98,053	30.51%	21.4%	24.81%	26.45%
Linear – Ridge	0.54	$96,896	33.67%	20.43%	19.75%	32.87%
Linear – Lasso	0.67	$99,996	32.78%	22.14%	23.82%	26.83%
Random Forest	0.66	$88,425	28.62%	16.90%	24.38%	30.04%
XGBoost	0.68	$82,132	27.24%	15.57%	21.00%	30.78%
Simple Neural Network	0.70	$80,114	22.7%	18.08%	22.86%	23.51%

* MAE = Mean Absolute Error

Modeling Summary: Choosing the Best Model

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  - Neural network was a simple 3-layer network

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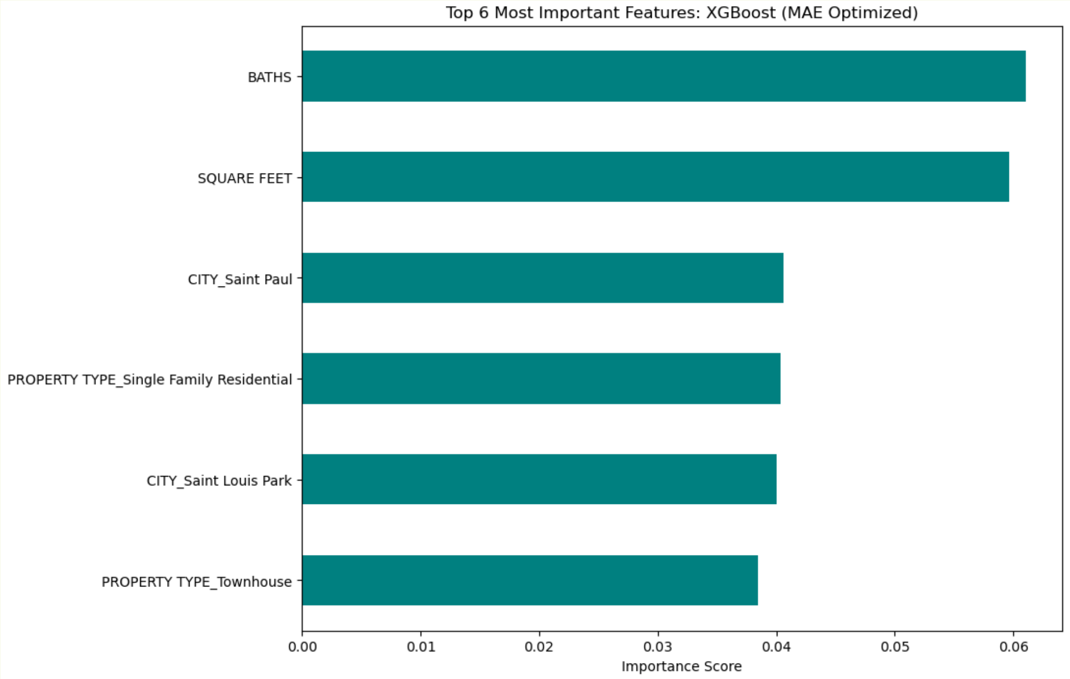
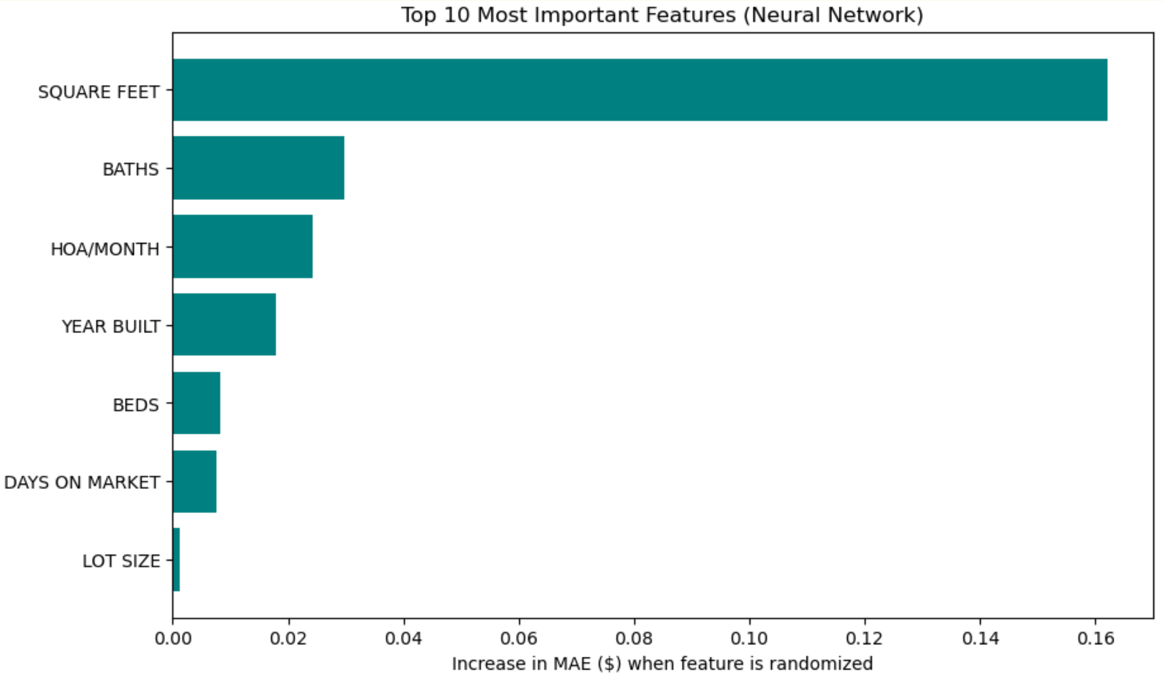
Modeling Summary: Feature Importance





Conclusions and Next Steps

- Conclusions
  - Depending on the price range, XGBoost or Neural Network would be the best models
  - Neural Network is the overall winner due to its consistency across price ranges
  - Ideally, these models would have less than 10% error
    - May need more data and feature engineering to make this happen
  - The models given here can give an idea as to whether a house is significantly over/under priced, but more refinement and data are needed to be more specific

- Next Steps
  - Obtain more data – more housing information would be very helpful and would allow for separating the houses based on price for modeling of separate bins
  - Gather more information about features
    - Features such as if there is a pool, granite countertops, newly remodeled, distance from schools, etc.
  - Further optimization of the Neural Network model
    - This was an initial Neural Network model that can be tuned further